

114–3海攬班高二上重補修_測驗

Topic 1: Advanced 1D Kinematics



Useful Formulas for kinematics

(1) $v = v_0 + at$

(2) $\Delta x = v_0 t + \frac{1}{2}at^2$

(3) $v^2 = v_0^2 + 2a\Delta x$

Q1: A bicycle's velocity increases steadily from 4 m/s to 16 m/s in 6 s.

(a) What is its average acceleration?

(b) Draw the Velocity–Time (v–t) graph for this bicycle. Calculate the displacement from $t = 0$ to 6 s from v–t graph.

Q2: A car accelerates from rest at a constant rate of 3 m/s².

(a) How much time does it take for the car to reach a final velocity of 15 m/s?

(b) Calculate the distance the car traveled during this time.

Q3: A train is traveling at 30 m/s. The engineer hits the brakes, and the train comes to a complete stop ($v = 0$ m/s) over a distance of 90 m.

(a) Calculate the train's constant acceleration.

(b) How many seconds it took for the train to stop.

Q4: A toy rocket is launched straight up into the air. Gravity acts on it with $a = -10$ m/s². It reaches a maximum height of 20 m before falling back down. At its maximum height, its instantaneous velocity is exactly 0 m/s.

(a) Calculate its initial launch velocity.

(b) Draw the Velocity–Time (v–t) graph for the rocket from launch until it reaches its highest point. Use the graph to find out how many seconds it took to reach the top.

Q5: A cyclist rides in a two–stage motion.

(a) Stage 1: She starts from rest and accelerates at 3 m/s²

- for 4 s. What is her speed at the 4-second mark? How far did she ride during Stage 1?
- (b) Stage 2: She then maintains that top speed (constant velocity) for another 5 s. How far did she ride during Stage 2?
- (c) Draw a v - t graph for this entire 9-second ride and calculate the total displacement via the graph.

Q6: Car A is driving at 30 m/s . Car B is 75 m directly ahead of Car A, driving in the same direction at a constant velocity of 10 m/s . The driver of Car A hits the brakes, creating a constant deceleration of -2 m/s^2 .

- (a) The danger of a collision only exists while Car A is moving faster than Car B. Calculate how many seconds it takes for Car A to slow down matching Car B's speed.
- (b) Using the time you found in (a), calculate the displacement of Car A and the displacement of Car B during this specific time period.
- (c) Assume Car A started at position $x_A = 0 \text{ m}$. This means Car B started at $x_B = 80 \text{ m}$. What are their final positions at time t ? Based on these positions, did Car A crash into Car B?
- (d) When they exactly crush to each other, which physical quantity of two cars will be the same? List the equation and solve for the exact second the crash happens.

Topic 2: 2D Kinematics and Projectiles



Useful Formulas for kinematics

(1) $v = v_0 + at$

(2) $\Delta x = v_0 t + \frac{1}{2}at^2$

(3) $v^2 = v_0^2 + 2a\Delta x$

Q1: You are standing on a bridge that is 80 m high. You throw a rock perfectly horizontally at a speed of 15 m/s . We use $g = -10 \text{ m/s}^2$.

- (a) What is its initial vertical velocity?
- (b) Calculate the time it takes to hit the water below.
- (c) Calculate the horizontal distance the rock traveled before hitting the water.

Q2: A snowboarder wants to slide off a perfectly flat horizontal ledge that is 45 m high. To win the competition, he must safely land on a specific snow pad that is exactly 60 m horizontally away from the base of the ledge.

- (a) Using the height of the ledge, calculate how long (t) the snowboarder will be in the air.
- (b) What must be the snowboarder's initial horizontal speed as he leaves the ledge to land exactly on the target?

Q3: A golfer hits a golf ball with an initial velocity of 50m/s at an angle of 37° above the ground.

- (a) Calculate the initial horizontal velocity (v_{0x}) and vertical velocity (v_{0y}).
- (b) Calculate the maximum height (H) the golf ball reaches.
- (c) Calculate the time that the golf ball takes to reach the highest point.

Q4: Based on the golf ball from Q3:

- (a) Find the total time the ball is in the air.
- (b) Calculate the total horizontal distance the golf ball traveled.

Q5: Based on the exact same golf ball from Q3 and Q4, and use the symmetry of projectiles:

- (a) Right before it hits the ground, what is its horizontal velocity (v_x) and vertical velocity (v_y) respectively?
- (b) Using your answers in (a) and the Pythagorean theorem, calculate the total speed of the ball right before it hits the ground.

Q6: A catapult fires a heavy stone with an initial velocity of 50 m/s at an angle of 53° above the ground. There is an enemy castle wall that is exactly 120 m away horizontally. The height of this castle wall is 60 m.

- (a) What is the horizontal and vertical velocity v_{0x} and v_{0y} ?
- (b) Calculate how many seconds (t) it takes for the stone to travel 120 m horizontally.
- (c) Calculate the height of the stone at that exact moment.
- (d) Did the stone fly over the wall? Explain your reason.